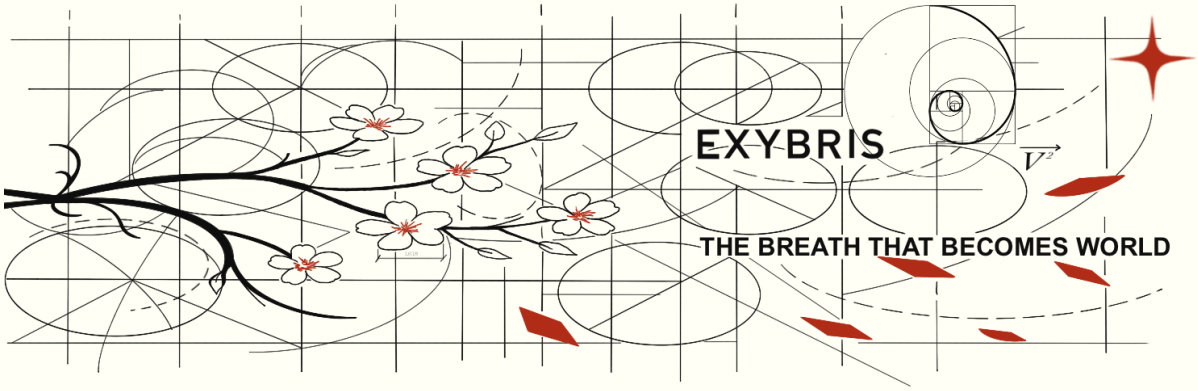

Research Note

Towards More Flexible Architectures: Bounded Endogenous Regulation as a Principle of Robustness in AI

Abstract

Contemporary AI architectures often maximize performance through specialization, compression, and rigidification of internal mechanisms. While this can improve certain benchmark scores, it may also increase system fragility in the face of context changes, non-stationary signals, perturbations, and internal drift.

This note argues for the opposite hypothesis: a degree of structural flexibility is not a computational deficiency, but a condition for proper functioning. By "flexibility", we mean the capacity of a system to adjust its internal dynamics in a contextual, continuous, bounded, and measured way, without losing its coherence.

The FPS framework (Fractal Pulsating Spiral) explores this direction through a metastable oscillatory system with endogenous regulation. The goal is not merely to propose a new theoretical object, but to test a more general idea: more flexible internal mechanisms can produce more robust representations, less fragile attention, and better adaptation to variable regimes. At this stage, FPS is still primarily validated within its own simulation framework, with an explicit roadmap towards AI applications such as self-regulated reservoirs and attention modulation.

Problem

A significant part of modern AI relies on powerful but relatively rigid mechanisms: fixed activation rules, poorly self-regulated internal dynamics, and adaptation often deferred to offline learning rather than the system's own dynamics.

This rigidity can be effective in stable settings, but becomes costly when:

- the signal changes regime,

- the input is noisy or irregular,
- multiple objectives must be arbitrated,
- local stability conflicts with global adaptation.

The question is therefore not only "how to increase performance", but:

how much flexibility must a system retain to continue functioning well when the context deforms?

Hypothesis

H1. An architecture with bounded, contextual endogenous regulation can better preserve its functional stability than a more rigid architecture subjected to non-stationary inputs.

H2. This internal flexibility can improve not only robustness, but also the quality of produced representations, by reducing unnecessary oscillations, dynamic collapses, and certain forms of excessive variance.

Intuition

A good system is not merely one that holds.
It is one that adjusts without breaking.

From this perspective, flexibility is not a concession to approximation; it is a structurally richer form of dynamic discipline. A system that is too rigid may appear performant because it strongly constrains its behavior, but this constraint may mask a deeper difficulty in absorbing real-world variability.

Proposal

The FPS program explores this idea through a network of metastable oscillators with endogenous regulation, positioned between descriptive model and prescriptive mechanism. The current framework operates on non-stationary signals and targets several internal performance metrics, with a separation between observable global signal, perceptual prior, and regulation modulation. The roadmap explicitly includes extensions towards:

- a self-regulated reservoir,
- an attention modulator for small-scale transformers,
- then oscillatory networks where regulation is integrated into the unit itself.

Testable Prediction

If this hypothesis is correct, a minimal AI version inspired by FPS should show, compared to a more rigid baseline:

1. less collapse in certain internal dynamics,

2. lower variance under perturbation,
3. better performance on non-stationary inputs,
4. an explicitly measurable trade-off between performance and robustness.

Position

This note does not claim that a flexible architecture will immediately outperform all dominant architectures. It supports a more precise proposition:

the excessive suppression of flexibility may constitute a design error, and the reintroduction of bounded contextual regulation could become an important principle for robust AI systems.

Next Step

The next step is not a total theory, but a simple and falsifiable experiment:

- implement FPS as a minimal module,
- choose a clear baseline,
- measure robustness, variance, stability and performance,
- demonstrate a clean effect, even a local one.

From Resilience to Internal Consideration

Initial results obtained with FPS suggest the existence of a non-trivial, stable, and reproducible dynamical regime. The observed visualizations correspond neither to trivial total synchronization nor to pure disorganization. They rather reveal a form of organized coherence, where strong local coherence can coexist with global heterogeneity — a non-zero but structured diversity. This configuration evokes a chimera-like regime, in the sense that the system seems to hold without fully homogenizing its components.

This point is important because it already supports a central intuition of the program: a system's ability to hold does not necessarily require maximal rigidification. A system can maintain order without crushing all its diversity. At this stage, FPS appears to produce not a simple lock-in, but a particular dynamic compromise between cohesion, variability, and stability.

Preliminary comparisons with control groups support this view. In the current experimental framework, FPS achieves a higher overall score than both controls (0.714 versus 0.448 for Kuramoto and 0.286 for Neutral — run_20260407_173917) and exhibits a distinct behavioral profile, marked by strong synchronization, good stability, high continuous resilience, and high fluidity, at the cost of higher computational expense and lower scores on certain dimensions such as innovation. This is interesting precisely because it does not produce a "perfectly winning" picture: it suggests less a miraculous system than one oriented towards a clean trade-off.

Beyond this comparative question, a deeper observation emerges from runs under Gaussian noise. Under variable perturbation, FPS appears to maintain high local coherence without manifest collapse, but at the cost of a significant increase in regulatory effort. This holding under constraint is an encouraging result: it suggests that the system's stability is not purely rigid, but actively maintained. However, it also opens a more demanding question.

Because a system can be stable, performant, and resilient while still relying on chronic compensation, over-effort, or forced maintenance. Yet this is not exactly what we seek here. The goal is not merely to produce an architecture that holds; it is to produce an architecture that holds in a way that is fair to itself.

We therefore propose to introduce a distinction between two regimes:

- **stability under constraint**: the system remains coherent, but at the cost of high effort, frequent mobilization, slow return to calm, or unequal distribution of the regulatory burden
- **stability with consideration**: the system remains coherent while better respecting its internal dynamics, with measured, reversible, non-punitive regulation, and without excessive suppression of local diversity

From this perspective, robustness alone no longer suffices as a criterion. One must also ask: at what internal cost is stability achieved? This question, already addressed in biology (allostasis vs. homeostasis, allostatic load), calls for a new family of metrics that could be grouped under the name **internal consideration metrics**: average regulatory load, return-to-calm time after perturbation, fraction of time in over-mobilization, cost distribution across strata, coherence per unit of effort, preservation of heterogeneity, or flexibility of de-escalation when constraint disappears.

Current results do not yet demonstrate this internal consideration, but they allow for a more refined reformulation of the research program. They suggest that FPS already knows how to respond, to hold, and to preserve its flexibility. The next step is to determine whether this holding can become more economical and distributed, more respectful of its own dynamics, and free from effort saturation.

In the FPS program, a good architecture should not merely maximize external performance. It should also minimize the regulatory violence required for its own maintenance.

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Exybris — Andrea Gadal | github.com/Exybris/FPS-real-tests_fractal-pulsating-spiral